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Exercise Physiology

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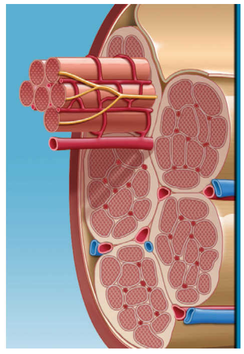

Exercise physiology is the identification and study of the physiologic mechanisms underlying physical activity and the body's response and adaptation to exercise. Most training programs used by athletes, whether they are for strength or endurance, aim to change the underlying physiology for the benefit of sport competition or fitness. This chapter discusses the major concepts of exercise physiology to provide a background that sports medicine practitioners can draw upon when treating or training athletes.

SKELETAL MUSCLE PHYSIOLOGY

The human body contains three types of muscle: smooth, cardiac, and skeletal. Training adaptations occur across all three types of muscle; however, this section focuses on skeletal muscle. Skeletal muscles support the body by working together to generate force across joints, leading to joint motion. Muscles also work to attenuate external forces in an effort to absorb shock and protect the joints.

Skeletal muscles are made up of different fiber types that have different properties, such as rate of contraction, force generation, and resistance to fatigue (Table 6.1). Skeletal muscle is made up of different fiber types. Fibers are currently classified by the myosin heavy chain isoform: one slow (type I) and three fast (IIA, IIB, and IIX). During the past several years, significant progress has been made in identifying signaling pathways that control the expression of muscle fiber types. With the recently discovered microribonucleic acids encoded within myosin genes that regulate muscle gene expression and performance,[1] a new individualized genetic differentiation may be possible.[2]

Muscle fibers are currently classified as types I, IIA, IIB, and IIX. In general type I muscle fibers operate at a slower twitch rate and are responsible for longer-term, endurance-type activities because of their ability to resist fatigue. Type IIA and IIB/X muscle fibers operate at a faster twitch rate and are responsible for high-force contractions, which dominate during explosive power maneuvers.

The true functional unit of the neuromuscular system is the motor unit. A motor unit is an alpha motoneuron (originating in the spinal cord) and all the muscle fibers that it innervates. Singular motoneurons innervate muscle fibers of the same contractile type in an all-or-none fashion. Although a muscle will contain more than one or all types of motor units, the percentage of each fiber type within a muscle contributes to its function and fatigability. Motor units can also be classified as either slow- or fast-twitch types, with the fast-twitch units divided into fatigue-resistant, fatigue-intermediate, and fatigable units. However, motor units can change in size in response to training and can also convert from one type to another.[3] This plasticity of both contractile and metabolic properties allows the skeletal muscle system to adapt to different functional demands.

Structure of a Skeletal Muscle

Bundles of fascicles encased in connective tissue (epimysium) make up skeletal muscles. Each fascicle is surrounded by a layer of connective tissue (perimysium), which contains multiple muscle fibers. The muscle fiber, itself an individual skeletal muscle cell, is cylindrical in shape, multinucleated, and composed of bundles of myofibrils surrounded by an endomysium (Fig. 6.1). These myofibrils contain thin and thick proteins that form repeating light and dark bands along the length of the myofibril, which give the muscle its striated appearance (Fig. 6.2); these repeating sections are called *sarcomeres* and are arranged end to end within each myofibril.

Sarcomeres are the functional, contractile units of skeletal muscle. Contractions are mediated through a dynamic interaction between actin and myosin, which are the contractile proteins. Spherical actin molecules in a helical arrangement form the thin protein structural lattice of the sarcomere. Each actin molecule contains a binding site protected by tropomyosin, a thread-like protein, and troponin, which acts as a stabilizing protein. Myosin molecules group together, each with a globular end, and form cross bridges, which are staggered toward opposite ends of the thick filament. These myosin molecules have a binding site with a high affinity for actin. This orientation of thick filaments at the center of the sarcomere provides the framework for sarcomere shortening and ultimately muscle contraction.

Physiology of Skeletal Muscle Contractions

Skeletal muscle contractions are a mechanical process driven by a neurochemical cascade. Several theories have been proposed about muscle contraction. The most widely accepted theory is the "sliding filament theory" proposed by Huxley and Hanson,[4] in which the active shortening of the sarcomere and the muscle results from actin and myosin "sliding" past one another while retaining their original length. An action potential from the motoneuron is propagated and acetylcholine is released from the axon at the neuromuscular junction, increasing permeability to sodium and potassium ions and causing an end-plate potential.

Abstract

Exercise physiology is the identification and study of the physiologic mechanisms underlying physical activity and the body's response and adaptation to exercise. Most training programs employed by athletes, whether they are for strength or endurance, aim to change the underlying physiology for the benefit of sport competition or fitness. This chapter discusses the major concepts of exercise physiology to give the sports medicine practitioner a background on which to draw from when treating or training athletes.

Keywords

cardiorespiratory
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mitochondrial adaption
muscle
neuromuscular response
Vo_2 max

TABLE 6.1 Classification of Human Skeletal Muscle Fibers

	I	IIa	IIx	IIb
Contraction time	Slow	Medium fast	Fast	Very fast
Size of motor neuron	Small	Medium	Large	Very large
Resistance to fatigue	High	Fairly high	Intermediate	Low
Activity	Aerobic	Long-term anaerobic	Short-term anaerobic	Short-term anaerobic
Mitochondrial density	High	High	Medium	Low

Modified from McArdle WD, Katch FI, Katch VL. *Exercise Physiology: Nutrition, Energy, and Human Performance*. 7th ed. Philadelphia: Lippincott Williams & Wilkins; 2009.

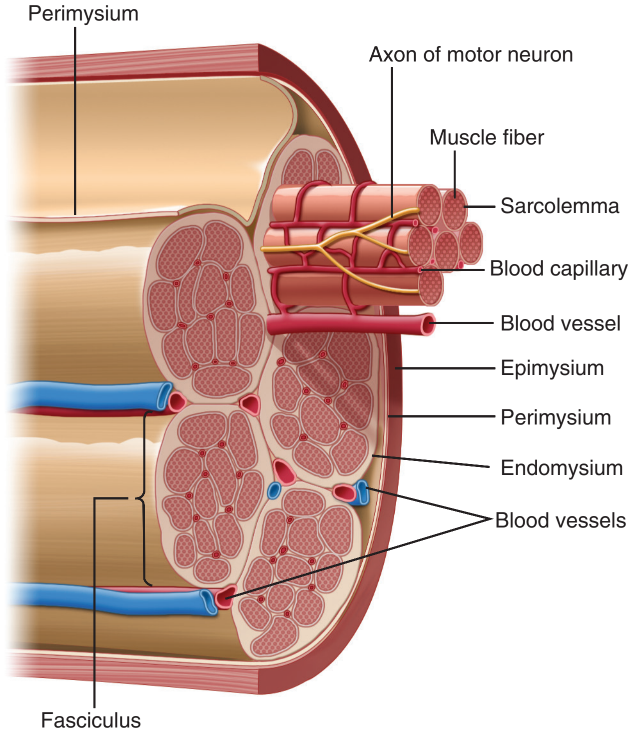

Fig. 6.1 Basic structure and organization of skeletal muscle with associated connective tissues, blood vessels, and motor neurons. (Modified from Palastanga NP, Field D, Soames R. *Anatomy and Human Movement—Structure and Function*. Edinburgh: Butterworth Heinemann; 2006.)

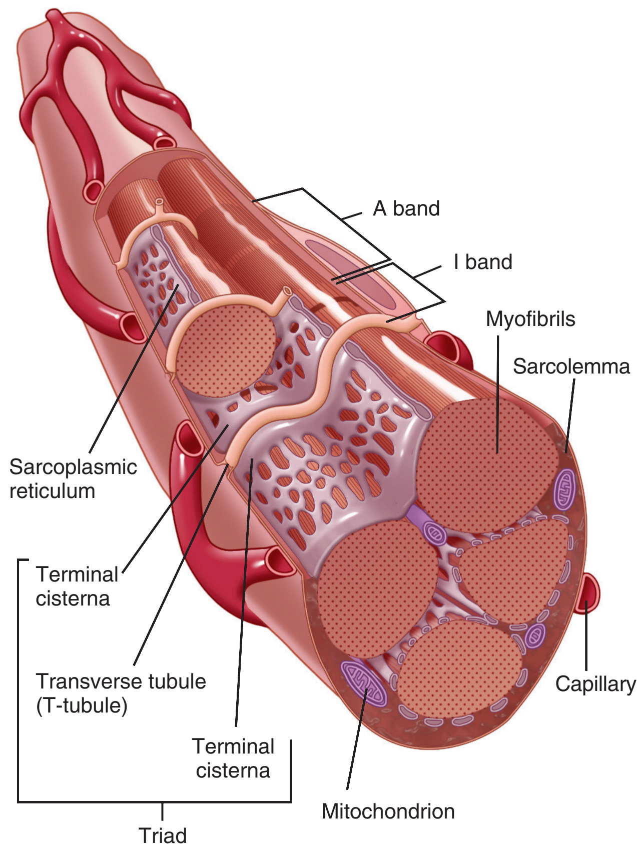

Fig. 6.2 Illustration of repeated sarcomere structure and organization within each individual muscle fiber. Each sarcomere contains thick (myosin) and thin (actin) contractile protein filaments that are responsible for sarcomere shortening. Alternating light and dark bands characterizing skeletal muscle are due to repeated bands of thick *(A band)* and thin *(I band)* filaments. The transverse (T)-tubule system is contiguous with the sarcoplasmic reticulum and sarcolemma. Once an action potential from a motor neuron is initiated, it propagates along the sarcolemma and is internalized to the myofibrils within a muscle fiber through the T-tubule system. (Modified from Seeley RR, Stephens TD, Tate P. *Anatomy and Physiology*. 3rd ed. St. Louis: Mosby; 1995.)

This potential travels along the muscle cell membrane through the transverse tubule system throughout all the myofibrils. Calcium ions released from the sarcoplasmic reticulum quickly bind to troponin molecules on the thin contractile filament, resulting in the binding of tropomyosin to the exposed binding sites on actin, which allows the myosin cross bridges to bind with actin, pulling the actin filaments toward the middle of the sarcomere (Fig. 6.3) and resulting in the filaments sliding past one another, causing muscle shortening and contraction. The collective shortening of myosin throughout the muscle is vital to force production and is termed the *power stroke*.

Calcium and adenosine triphosphate (ATP) are cofactors (nonprotein components of enzymes) required for the contraction of muscle cells. ATP supplies the energy, as previously described, but what does calcium do? Calcium is required by two proteins, troponin, and tropomyosin, that regulate muscle contraction by blocking the binding of myosin to filamentous actin. The power stroke requires the hydrolysis of ATP, which breaks a high-energy phosphate bond to release energy, leading to the release of the linkage between myosin and actin to prepare the myosin for another power stroke or for relaxation. If no new ATP is available, this linkage cannot separate, resulting in rigor mortis. Power-stroke cycles continue until the concentration of calcium is no longer sufficient to cause the binding of actin and myosin.

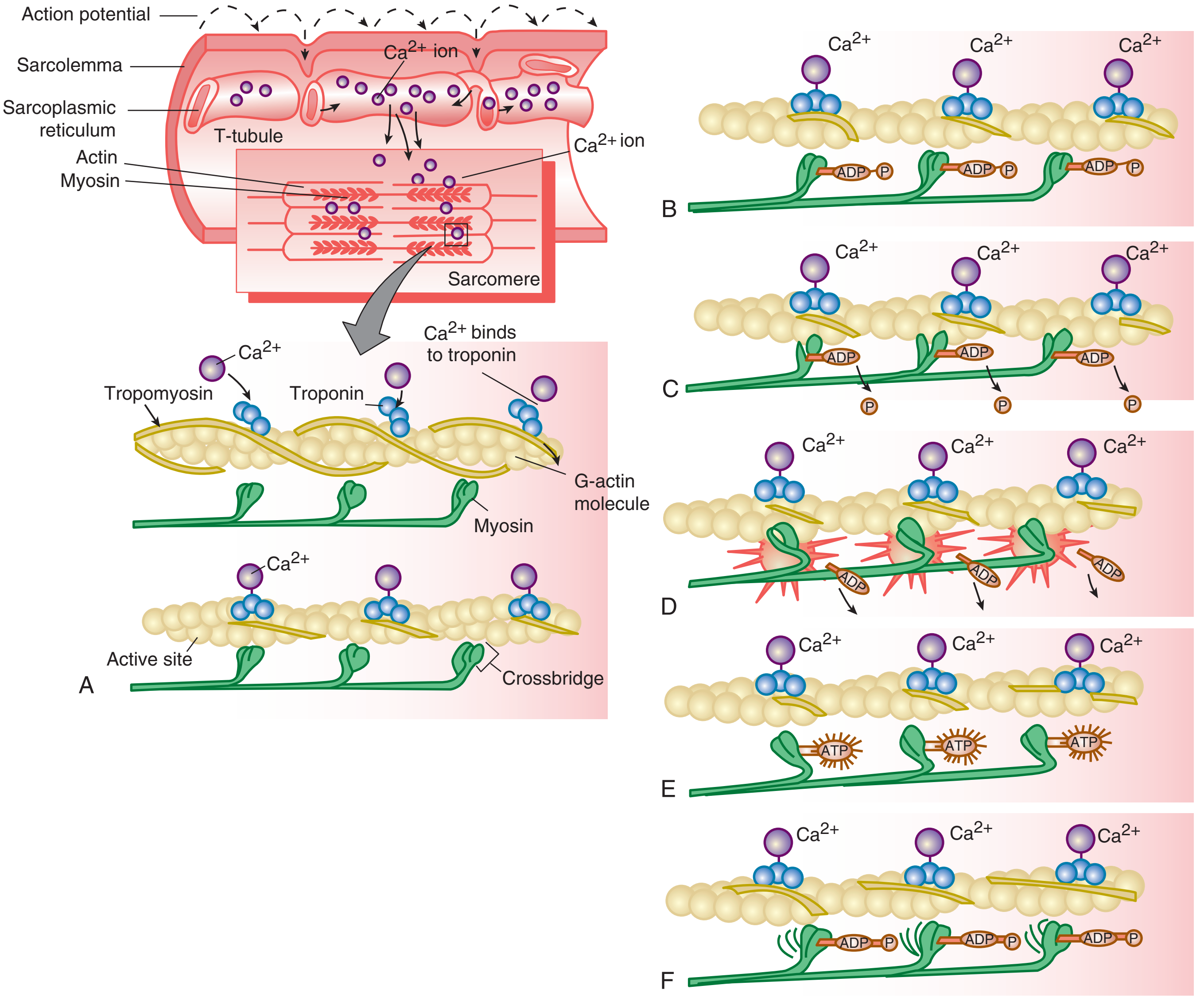

Fig. 6.3 Sequence of skeletal muscle contraction: sarcolemma depolarization causes calcium release from the sarcoplasmic reticulum. (A) Calcium binds with troponin and shifts tropomyosin molecules to expose myosin-binding sites on actin. Myosin cross bridges bind to actin, producing a "power stroke" of contraction. Adenosine triphosphate *(ATP)* is needed to break the link and prepare for the next cycle. Cycles (B to F) continue as long as sufficient calcium is present to inhibit the troponin-tropomyosin system from blocking actin-binding sites. *ADP,* Adenosine diphosphate; *P,* phosphate; *T-tubule,* transverse tubule. (Modified from Seeley RR, Stephens TD, Tate P. *Anatomy and Physiology.* 3rd ed. St. Louis: Mosby; 1995.)

Since the publication of the sliding filament theory by Huxley and Hanson in 1954,[4] new mechanisms have been proposed for muscle contractions. Several questions still remain, including enhancement of contractile force with stretch, depression of force with muscle contraction, and efficiency of force production during contraction. The winding filament model[5] and the sticky spring model[6] may both provide a means to assist in answering these questions by acknowledging the importance of the protein titin. Titin is activated by calcium influx and is wound upon the thin filaments by the cross bridges, which shorten but also rotate the thin filaments. Titin is also actively involved in force regulation of skeletal muscles, especially when muscles are stretched actively to long sarcomere lengths.[7] These new models account for force enhancement during stretch and force depression during shortening, but additional study is needed to further validate these theories.

Force Production of Muscle

Muscle force can be increased or decreased by the recruitment of more or fewer motor units. For any motor task, many possible combinations of motor units can be recruited, and it has been proposed that a simple rule, the "size principle,"[8] governs the selection of motor units recruited for different contractions. Motor units can be characterized by their different contractile, energetic, and fatigue properties, and it is important that the selection of motor units recruited for given movements allows units with the appropriate properties to be activated.[9] The structural presence of titin also plays a role in the force generated by muscle.[10]

Types of Muscle Contractions

Skeletal muscle contractions produce force and control joint movement and body control; they are capable of different activities and demands through different types of muscle contraction. Isometric contractions occur when force is generated but no apparent change in total muscle length occurs. Concentric muscle contractions generate a force that shortens the muscle. Eccentric muscle contractions occur when a force is generated as the muscle lengthens. Isotonic contractions produce muscle tension and joint movement against a constant load with a variable rate of movement. Isokinetic contractions have a constant rate of movement, which is maintained by varying the amount of muscle effort. Isokinetic efforts are rare in sport training but are utilized in the rehabilitation setting. In weight lifting or sport training, the two most common types of muscle contractions are the isometric, concentric activities, where muscles are shortened in a voluntary fashion to lift a constant load (weight), and the eccentric contractions, where the muscle lengthens against a constant load (negatives).

Although concentric muscle contractions are the contractions most people consider, eccentric contractions are probably more useful in the sports training and rehabilitation setting. During locomotion, eccentric contractions act to dissipate energy both as heat and as a shock absorber, reducing kinetic energy through a braking effect.[11] The energy absorbed within the stretched muscle-tendon can then be stored as potential energy and returned, allowing the muscle-tendon complex to act as a spring.[12] The muscle protein titin may be important in this spring-like property of the muscle.[13] Finally, the energy cost for eccentric contractions is unusually low,[14] whereas the magnitude of forces produced is very high. Therefore as a response to eccentric loading, muscles gain strength, which makes eccentric loading well suited for both strength training and as a rehabilitation tool.

Energy Metabolism of Skeletal Muscle

Exercise results in an increased rate of ATP demand. To sustain muscle contraction, ATP must be regenerated at a rate complementary to ATP demand. Three energy systems (phosphagen, glycolytic, and mitochondrial respiration) are available to synthesize ATP in muscle (Fig. 6.4). The three systems differ in the substrates they use, the products they generate, their maximal rate of ATP regeneration, and their capacity for ATP regeneration.

Phosphagen Energy System

The phosphagen system is important for exercises that require short-term singular muscle contractions or a limited number of intense, near-maximal muscle contractions. Within the phosphagen system, the creatine kinase and adenylate kinase reactions both produce ATP, but the creatine kinase reaction has a much higher capacity for ATP regeneration, given the large store of creatine phosphate (CrP) at rest. Another important feature is the adenylate kinase reaction, which produces adenosine monophosphate (AMP), which then activates two enzymes

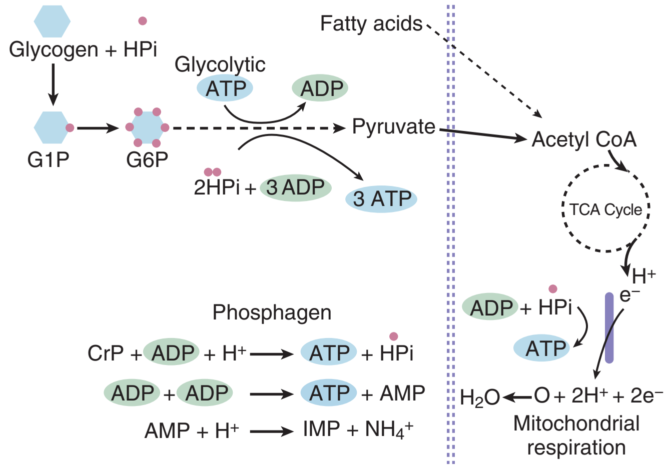

Fig. 6.4 Energy systems to synthesize adenosine triphosphate *(ATP)* in muscle. *ADP,* Adenosine diphosphate; *AMP,* adenosine monophosphate; *ATP,* adenosine triphosphate; *CrP,* creatine phosphate; *G1P,* glucose-1-phosphate; *G6P,* glucose-6-phosphate; *HPi,* hydrogen phosphate ion.

influential in glycolysis. AMP activates phosphorylase, which increases glycogenolysis and the rate of glucose-6-phosphate (G6P) production that becomes immediate fuel for glycolysis. AMP also activates phosphofructokinase, allowing an increase flux of G6P through glycolysis, which increases the rate of ATP regeneration.

It has been hypothesized that during the first 10 to 15 seconds of exercise, CrP is solely responsible for ATP regeneration.[15] Near-complete recovery of CrP may take from 5 to 15 minutes, depending on the extent of CrP depletion, severity of metabolic acidosis, and characteristics of the muscle fibers utilized.[16] Evidence is conflicting about the importance of oxygen during the resynthesis of CrP after high-intensity exercise. Some investigators suggest that CrP resynthesis is reliant on oxidative metabolism,[17] whereas others found that after high-intensity exercise under ischemic conditions, glycolytic flux remained elevated for a short period.[18] More recent work, however, supports the theory that glycolytic ATP production may contribute to CrP resynthesis during the initial fast phase of recovery after high-intensity exercise.[19]

Glycolytic Energy System

When exercise continues for longer than a few seconds, the energy to regenerate ATP is derived primarily from blood glucose and muscle glycogen stores. This prolonged exercise increases the production of AMP. This production, coupled with the increases in intramuscular calcium and inorganic phosphate levels, causes increases in the phosphorylase reaction and the increased uptake of glucose by active muscle. The increased rate of G6P production from glycogenolysis and the increased glucose uptake provide a rapid source of fuel for several pathways that degrade G6P to pyruvate within the glycolytic pathway.

Traditionally it was thought that CrP was the sole fuel used at the initiation of glycolysis, with glycogenolysis occurring only once the CrP levels began to fall. However, research suggests that ATP resynthesis from glycolysis during 30 seconds of maximal exercise begins almost immediately with exercise.[20,21] Maximal ATP regeneration capacity from glycolysis occurs when exercise at or above maximal oxygen uptake is performed for 2 to 3 minutes or as long as is sustainable by the athlete.[22]

Lactate. During higher-intensity exercise, an incomplete oxidation of glucose or glycogen occurs, with the excess pyruvate being converted to lactate via the lactate dehydrogenase reaction. For many years this leftover lactate was considered a waste product and a major cause of muscle fatigue or drop in performance.[23] However, starting in the 1970s and continuing with the recent work of Brooks,[24] this view has been challenged. It has now been shown that lactate is in fact beneficial during intense exercise. Production of lactate is essential in removing pyruvate to sustain glycolysis, and it is an oxidation-reduction process, reducing oxidized nicotinamide adenine dinucleotide (NAD+) to reduced nicotinamide adenine dinucleotide phosphate (NADPH), which allows glycolysis to continue to produce ATP. Without lactate this process would slow and the amount of ATP, and thus the energy available for muscles, would drop. Most of the lactate formed during steady-state exercise is removed by oxidation and only a small amount is converted to glucose. The lactate may also be "shuttled" from areas of high glycogenolysis to areas of high respiration via lactate shuttles.[25] Although production of lactate initially was thought to occur only when anaerobic work was performed, it has been demonstrated that lactate is a valuable fuel during oxidative metabolism as well.[26]

Mitochondrial Respiration (Oxidative System)

The connection between glycolysis and the mitochondria is complete when pyruvate and the electrons and protons from the reduction of NAD+ to NADH are transferred into the mitochondria as substrates for mitochondrial respiration. The resynthesis of ATP involves the combustion of fuel in the presence of sufficient oxygen and occurs in the mitochondria. The fuel can be obtained from within the muscles (free fatty acids and glycogen), outside the muscle (fats), and in the form of blood glucose (ingestion of carbohydrates or release from the liver).

Carbohydrates in the form of glucose are the major fuel of the body during exercise. Humans gain most of their glucose from their diet. In the resting state, glucose is stored in the form of glycogen in both the liver and in skeletal muscle. The main fatty acid utilized at rest and during muscle contractions is palmitate (a 16-carbon fatty acid). Because the inner mitochondrial membrane is impermeable to fatty acids larger than 15 carbons, the fatty acids are transported via the carnitine shuttle into the mitochondria. Finally, muscle has an ample supply of amino acids (free amino acids) that can be used in a catabolic state when carbohydrate supply is too low. However, if carbohydrate supply is inadequate, protein catabolism and amino acid oxidation must occur. Intense exercise also increases this amino acid oxidation. Carbohydrate oxidation is the most efficient fuel in the production of ATP.

The energy systems respond differently to the diverse demands of sporting exercise. The nonmitochondrial (anaerobic) system is capable of responding immediately and is able to support high-power output via extreme muscle force. The phosphagen system is utilized for explosive, maximal efforts, which are very short in duration. The glycolytic pathway is used for activities that require sustained, intense efforts but are not maximal. Unfortunately the anaerobic system is limited in its capacity, such that a reduction in power output will occur unless the demands can be supplemented by the aerobic system. The aerobic system is capable of responding quickly, yet it is incapable of meeting the initial high demands of intense exercise and is much better suited for longer efforts. The oxidative system is best for long-duration, low-intensity activities.

MUSCULAR RESPONSE TO TRAINING

Skeletal muscle exhibits superb plasticity in response to changes in functional demands. Chronic increases of skeletal muscle contractile activity, such as endurance exercise, lead to a variety of physiologic and biochemical adaptations in skeletal muscle, including mitochondrial biogenesis, angiogenesis, and fiber-type transformation. These mitochondrial adaptions can be increased through the use of high-intensity interval resistance training methods.[27] These adaptive changes are the basis for the improvement of physical performance and other health benefits.[28]

Muscles gain strength by getting larger (hypertrophy) and also by increasing skeletal muscle cell division (hyperplasia). The overload principle states that when a muscle is exposed to a stress or load that is greater than what it usually experiences, it will adapt so that it is able to handle the greater load.[29] A meta-analysis of 17 studies concluded that mechanical overload resulted in increases in muscle mass, muscle fiber area (hypertrophy), and muscle fiber number (hyperplasia).[30] Increases in fiber area were approximately twice as great as increases in fiber number.[31]

As a skeletal muscle hypertrophies, contractile proteins are synthesized, and the muscle is therefore capable of producing more force. In response to heavy resistance training, type IIA fibers exhibit the greatest growth, whereas types IIB and I exhibit the least amount of growth.[32] Muscle hypertrophy is more common in fast- than in slow-twitch muscles. Strength training leads to muscle hypertrophy, which increases muscle mass and typically occurs after weeks of such training. It appears that hyperplasia in skeletal muscle is greatest when certain types of mechanical overload, particularly stretch, are applied.

Muscle fibers within a motor unit have the same fiber type; however, skeletal muscles may contain several types of fibers based on demand of the muscle. This fiber type is genetically determined, but it may be altered with a training stimulus. Although the type of fiber cannot be changed, the amount of area taken up by a fiber type can be altered with training. Heavy resistance training causes increased type IIA and decreased type IIB fibers, whereas type I fiber composition in human skeletal muscle remains unchanged.[33] In bench studies, structural and genetic characteristics of muscle fiber types have been modulated with fiber-specific stimulation[34]; it is unclear if this modulation will translate to human skeletal muscle.

Many of the initially observed strength gains of a weightlifting program are mostly due to neuromuscular adaptations.[32] As exercise intensity increases and muscles begin to fatigue, the nervous system recruits larger motor units with higher frequencies of stimulation to provide the force necessary to overcome the imposed resistance. Early strength gains and increased muscle tension production from training result from a more efficient neural recruitment process.[35,36]

In recent years, however, research has demonstrated that traditional strength training methods may be augmented through use of low-load resistance training with blood flow restriction (LL-BFR). The use of LL-BFR training with low loads has been found to yield hypertrophy responses comparable to those found with heavy-load resistance training.[37] It is hypothesized that an ischemic and hypoxic muscular environment is generated during blood flow restriction, which causes high levels of metabolic stress. This metabolic stress combined with the mechanical tension of training is theorized to activate factors associated with muscle growth. These factors may include systemic hormone production,[38] production of reactive oxygen species,[39] anabolic/catabolic signaling,[40] and/or increased fast-twitch muscle recruitment.[41] Currently, however, these factors are still only theorized and more research is needed to better elucidate the exact mechanisms.[42]

MUSCLE FATIGUE

Muscles that are used intensively show a progressive decline of performance that largely recovers after a period of rest; this reversible phenomenon is denoted as *muscle fatigue*. Many muscle properties change during fatigue, including the action potential, extracellular and intracellular ions, and many intracellular metabolites. Fatigue can be the result of peripheral causes, at the level of the skeletal muscle, or central causes.

Energy consumption of skeletal muscle increases dramatically with high-intensity exercise. The rate of energy need can exceed the aerobic capacity, forcing the muscles to rely on the anaerobic pathways for energy. Because high-intensity exercise leads to fatigue, it seems plausible that a relationship may exist between anaerobic exercise and fatigue. Anaerobic exercise leads to an accumulation of inorganic acids, mostly lactic acid, which dissociates into lactate and hydrogen ions.[43] As previously discussed, lactate is a muscle fuel and therefore is probably not a factor in muscle fatigue, but the increase in hydrogen ions may be an important cause of muscle fatigue. Studies supporting this theory have shown a good temporal correlation between the decline of muscle pH and the reduction of force or power production.[44] In addition, studies have demonstrated that acidification may reduce both the isometric force and the shortening velocity.[45]

Peripheral fatigue can be a result of a reduced level of calcium during contractions, or it may be related to the contractile proteins themselves. During exercise, muscles depend on an ATP pump to return calcium to the sarcoplasmic reticulum; as exercise continues and the level of ATP drops, the ability to return calcium back to the sarcoplasmic reticulum drops and peripheral fatigue results. Another mechanism of peripheral fatigue occurs at the level of the cross bridges. It has also been demonstrated that the initial fall in muscle force is due to a reduction of the force generated by the individual cross bridges, whereas as fatigue progresses, a decline in the total number of cross bridges actually occurs.[46] Finally, when muscles are fatigued, fewer motor units are available; thus reduced motor unit activation is a cause of fatigue.[47]

Exercise performance is also dictated by the perception of effort of the brain and its inhibitory effects on central motor drive to the periphery. It is believed that these peripheral and central factors are closely linked, with the development of peripheral fatigue influencing the rate of development of "central" fatigue, operating through neural feedback pathways from working muscle to the motor control areas of the central nervous system.[48,49]

Differentiation between central versus peripheral etiologies of muscle fatigue can be investigated via electrical stimulation. Muscles can be stimulated in a nonfatigued state and then again while fatigued, and if the characteristics of the contraction are different before and after fatigue, the likely cause is peripheral. If the contraction is reduced before fatigue, then the cause may be more central.[50,51] Other etiologies of central fatigue are psychological and may be more difficult to determine. A major direction for future studies should be to identify the mechanisms that contribute to human fatigue in various activities and particularly during disease processes.

NEUROMUSCULAR ADAPTATION TO EXERCISE

Neuromuscular performance is determined not only by the quality and quantity of involved muscles but also by the ability of the nervous system to appropriately activate the muscles. These neural adaptive changes in response to training are referred to as neural adaptation. These neural factors are important for increases in strength early in resistance training, whereas muscular hypertrophy of trained muscles develops after prolonged resistance training.[52] In order for athletes to make performance gains, neuromuscular adaptation must occur.[36]

Resistance training–induced adaptations are supraspinal in nature and include increased excitation[53] and changes in the organization of the motor cortex.[54] Such adaptations can influence the manner in which muscles are recruited for functional tasks. Changes at the neuromuscular junction, which are associated with functional alterations in transmission, may lead to enhanced neuromuscular transmission. These adaptations can improve the activation of muscles and are likely the result of muscle training.[54] Increased levels of muscle activation and subsequent increase in force achieved by increases in individual firing at the motor unit leads to the recruitment of more motor units.[55]

In addition to these neural factors, the amount of force that a muscle can exert is influenced by the number and size of fibers and the fiber type. Heavy resistance exercise results in increases in strength and changes in both neuromuscular function and muscle morphology.[56,57] Finally, neural adaptations are found in the level of coactivation of antagonist muscles[58] and changes in the synergistic patters of muscle activations,[59] which could also contribute to maximization of force generation.

In contrast to resistance training, the neural response to endurance training involves strategies to improve efficiency, such as reducing the number of motor units needed to maintain force and increasing the activation of synergistic muscles.[60] Endurance training results in increased capillary density and decreased muscle cross-sectional area to facilitate oxygen delivery to tissues. In addition, endurance training leads to an increase in cellular mitochondrial content and oxidative enzyme activity as well as increased aerobic capacity. These changes require 8 to 12 weeks of training to be realized. The net results of long-term endurance training are increased metabolic capacity of muscle, increases in muscular force output, and muscle hypertrophy.

DELAYED-ONSET MUSCLE SORENESS

The demands of the training stimulus causes microdamage to muscles, and their ability to recover and regenerate is part of the training process. Muscle soreness may occur during this period, which peaks 24 to 48 hours after a strenuous bout of training and typically resolves by 96 hours. This phenomenon is known as *delayed-onset muscle soreness* (DOMS). DOMS is typically experienced by all persons regardless of fitness level and is a normal physiologic response to increased exertion and the introduction of unfamiliar physical activities. The resulting sensation of pain and discomfort can impair physical training and performance; thus the etiology and prevention of DOMS is of interest to athletes and trainers.

More muscle damage may occur with higher-intensity or unfamiliar exercises, predominantly eccentric loading exercises. Other factors that may be involved include muscle stiffness, contraction velocity, fatigue, and angle of contraction.[61] Some evidence indicates that fast-twitch muscle fibers are more susceptible to eccentric-induced injury.[62] The initial injury to the muscles is a mechanical disruption of the sarcomeres,[63] which proliferates a secondary inflammatory response[64] releasing prostaglandin E2 and leukotrienes, giving the sensation of pain and contributing to swelling as a result of increased vascular permeability. The initial injury pattern is sporadic throughout the muscle, and the damage to the sarcomeres does not extend to the length of the myofibril or across a whole muscle fiber.[62] Thus the muscle injury pattern of DOMS differs from that of an acute muscle strain, which occurs as an isolated disruption of the muscle-tendon junction extending across the fibers.[65]

Can the symptoms and functional consequences of DOMS be mitigated or prevented? It would seem reasonable that if DOMS is thought to be an inflammatory problem and is mediated by prostaglandins, the use of nonsteroidal antiinflammatory drugs (NSAIDs) or the application of ice would help to reduce or prevent DOMS. However, studies show mixed results with NSAIDs,[66,67] and because NSAIDs are not without risk, further data are needed to recommend the use of NSAIDs to treat DOMS. Ice offers an analgesic effect but little sustained benefit.[68] Muscle stretching does not appear to produce clinically important reductions in DOMS in healthy adults[69]; however, foam rolling may.[70] The ideal strategies (time, amount of pressure applied during foam rolling, etc.) are not known at this time.[71] Antioxidants are not effective and may actually delay muscle recovery.[61] However, studies do show that massage,[72–74] as well as massage plus the use of compressive clothing,[75] is of benefit in reducing the symptoms of DOMS. Finally, whole-body vibration prior to exercise may be effective for reducing DOMS.[76]

STRENGTH TRAINING IN YOUNG ATHLETES

Strength training in young athletes has been controversial. Some of the controversy originated from misconceptions about benefits of training and the risk to the skeletally immature. Previously it was thought that prepubertal athletes could not benefit from weight training because of their low levels of gonadal hormones[77]; it was also feared that youths might lose flexibility and range of motion with strength training.[78] Finally, it was thought that strength training would expose the growth plates of young athletes to excessive risk of injury.[77] Recently, however, research has demonstrated that well-designed resistance training programs can enhance the strength of children and adolescents beyond what is expected with normal maturation.[79] Many leading sports organizations now recognize the importance of muscular strength during childhood and adolescence[80–83] both to improve performance and for injury prevention.[84]

Strength gains of up to 75% can be observed in young athletes after an 8- to 12-week training program.[83] Lighter loads and a greater number of repetitions (10 to 15) are recommended to increase strength.[85,86] These training-induced changes in youths are predominately due to neural adaptations as opposed to the muscle hypertrophy seen in adults.[85] Strength training in the young also has the potential to offer other many health benefits. Improved bone health,[87] body composition, reduction of cardiovascular risk factors,[88] and decreased risk of sports injury[89] result from strength training. Therefore it would appear that the benefits outweigh the risks and that properly designed and supervised programs are safe and effective for young athletes.

EFFECTS OF AGING ON SKELETAL MUSCLE

Aging has been associated with loss of muscle mass, which is referred to as *sarcopenia*. This decrease begins around age 50 years but becomes more dramatic after 60 years.[90] Aging negatively affects power, strength, and endurance in skeletal muscles. Recently muscle power has been found to be a better predictor of functional performance than muscle strength.[91] Loss of power contributes to a decrease in short-term anaerobic performance and is directly related to the increase in falls among older people and impairment in the performance of daily activities.[91] The etiology of this loss of power is from both a loss of muscle mass and a slowing in the firing of motor neurons[92] and maximal shortening velocity.[93] The loss of strength is directly related to the loss of muscle mass. Sarcopenia accounts for more than 90% of age-related strength decrease.[94] Other factors involved in strength reduction are decreases in the tension of muscle tissue due to aged myofibrils[95] or a decrease in the number of functioning cross bridges.[96] Despite the negative effects of aging, strength training can improve power and strength, reversing some of the decrement and leading to improved functionality in older adults. Despite the loss of muscle mass, the addition of high-intensity interval training (HIIT) increases the mitochondrial content almost back to that of younger athletes.[97] High-velocity resistance training[98,99] and increasing lower extremity strength[100] have been shown to be the best way to achieve this goal.

TABLE 6.2 **Hormonal Responses to Exercise**

Hormone	Change	Function
Pituitary		
Growth hormone	Increases proportionally to exercise intensity	Stimulates metabolism
β-Endorphins	Increase with exercise above 60% $\dot{V}O_2$ max	Analgesic, "natural high"
Antidiuretic hormone	Increases with exercise	Maintains hydration
Prolactin	Increases with exercise	Unclear
Adrenal		
Glucocorticoids, adrenocorticotropic hormone	Increase with exercise above 50% $\dot{V}O_2$ max	Stimulate metabolism
Aldosterone	Increases proportionally to exercise intensity	Maintains hydration
Epinephrine and norepinephrine	Increase proportionally to exercise intensity[a]	Mediate cardiovascular responses
Pancreatic		
Insulin	Decreases proportionally to exercise duration (and blood glucose levels)[a]	Guards against hypoglycemia
Glucagon	Increases proportionally to exercise duration	Increases serum glucose levels
Gonadal		
Testosterone	Increases proportionally to exercise intensity[a]	Unclear
Estrogen and progesterone	Increase with exercise[a]	Unclear

[a]Basal levels also decrease with long-term endurance training.
$\dot{V}O_2$ max, Maximum oxygen uptake.

CORE STRENGTH AND NEUROMUSCULAR TRAINING

Core stability is an important component in maximizing efficient athletic function as well as in preventing injury. Core stability can be defined as the "ability to control the position and motion of the trunk over the pelvis to allow optimum production, transfer and control of muscle activity."[101] The core consists of the muscles of the trunk and pelvis as well as the neural, osseous, and ligamentous elements. Its stability is dependent not only on muscular strength but also on proper sensory input that alerts the nervous system about interaction between the body and the environment, providing constant feedback and allowing for refinement in movement.[102] These proprioceptive abilities have a positive impact on reducing injuries,[103] and exercise programs targeting the enhancement of these abilities may prevent injuries.[104,105] Some evidence indicates that balance training or multifaceted training programs are effective in preventing lower limb injuries such as ankle sprains[106,107] and anterior cruciate ligament tears[108–110] among young adults when they participate in ball sports.[111] Finally, after injuries have occurred, rehabilitation of deficits may prevent future injuries.[112] To date, however, the most effective training program and the optimal frequency and duration of these sessions have yet to be determined, and more research is needed.[113]

HORMONAL ADAPTATION TO EXERCISE

Many changes that occur in response to exercise are mediated by hormones. Growth hormone (GH), endorphins, and antidiuretic hormone from the pituitary assist in the performance and recovery of muscles. Adrenal hormones such as catecholamines, glucocorticoids, and mineralocorticoids are responsible for most of the key changes needed in exercise, although many other hormones are also involved. Pancreatic and gonadal hormones also play a key role (Table 6.2).

Pituitary

GH is a peptide hormone that stimulates muscle growth, cell reproduction, and cell regeneration. GH also stimulates protein synthesis and promotes lipid metabolism, both of which conserve blood glucose levels during exercise to allow for sustainment of performance and contribute to muscle regeneration. GH levels are increased with resistance exercise.[114] The exact mechanism is not known, but afferent stimulation, lactate, or nitric oxide may play a role.[115] Both concentric and eccentric loading resistance training exercises lead to increased GH levels.[116,117] Endurance exercise training above the lactate threshold may amplify the pulsatile release of GH at rest, increasing 24-hour GH secretion; however, studies suggest that weight training may lead to a higher level of serum GH.[118] Another potent stimulator of GH is sleep,[119] which underscores the importance of sleep in recovery for athletes in training.

Administration of supraphysiologic doses of GH to athletes as a performance enhancer may increase lean body mass and improve athletic performance.[120] However, the model of acromegaly demonstrates that long-term GH excess does not ultimately improve performance and may be dangerous. Recently a systematic review failed to find any evidence to show that exogenous GH administration improves sports performance.[121] However, despite the lack of scientific evidence, GH is banned by most sporting agencies.

β-Endorphins are endogenous opioid peptides secreted by the pituitary in response to exercise. Exercise of sufficient intensity and duration has been demonstrated to increase circulating β-endorphin levels. β-Endorphins are active at opioid receptors, provide analgesic effects[122] and may be responsible for the "runner's high" experienced by persons who perform vigorous endurance exercise.[123]

Muscle activity and exercise cause perspiration, which leads to fluid loss and hemoconcentration of electrolytes as a result of the relative loss of water and sodium. This low plasma volume stimulates the posterior pituitary to release antidiuretic hormone to promote water concentration by increasing the permeability of the kidneys' water-collecting ducts, leading to less water being excreted and minimizing the risk of dehydration. Exercise also leads to increased prolactin secretion from the pituitary,[124] which is maximized after a hard anaerobic effort.[125] The exact role of prolactin in response to exercise is unclear, but it has been shown to stimulate the phagocytic activity of the immune system,[124] possibly in an early attempt to repair the exercise-damaged muscles or as an early precursor to DOMS.

Adrenal Hormones

Glucocorticoids, primarily cortisol, are released from the adrenal cortex in response to pituitary adrenocorticotropic hormone. Cortisol functions to stimulate liver gluconeogenesis, adipose lipolysis, and protein degradation in both liver and predominantly type II muscle fibers.

Levels of both adrenocorticotropic hormone and cortisol have been shown to rise in response to resistance and endurance training. The minimum intensity of exercise (i.e., threshold) necessary to produce a cortisol response is 60% of the maximum oxygen uptake (VO_2 max). Above 60% VO_2 max, a linear increase between the intensity of exercise and the increase in plasma cortisol concentrations is observed.[126] Cortisol levels may not increase at exercise below this intensity and may in fact decrease.[127] Other factors that modulate cortisol release include hypohydration, meals, and time of day. Hypohydration (up to 4.8% of body mass) amplifies the exercise-induced response of cortisol to exercise.[128] Cortisol response to exercise is significantly higher during evening exercise compared with morning exercise.[129]

Finally, enhanced activation of the sympathoadrenal system may occur in game or race situations and may be one of the key "driving forces" to performance improvement.[130]

Aldosterone functions in the kidney to prevent sodium and water loss. Aldosterone secretion is regulated by the kidneys and is driven by decreased renal blood flow accompanying hypohydration and shunting of blood peripherally to lower temperature during exercise. Aldosterone levels increase in a graded fashion with hypohydration, heat stress, and exercise intensity.[131] An additive effect of combined hypohydration and exercise intensity is also believed to exist, and these responses are closely coupled to plasma osmolality.[131] Response can be significant, with marathon runners having documented increases in aldosterone levels for almost a full day after completion of a race.[132,133] This prolonged elevation in aldosterone levels may be important to athletes who sustain an injury that requires surgery shortly after prolonged competition.

Catecholamine (epinephrine and norepinephrine) secretion from the adrenal medulla increases with exercise. These hormones have a range of effects that include increased heart rate (HR), respiratory stimulation to meet aerobic demands, and generalized vasoconstriction with vasodilatory effects on skeletal and cardiac muscles to promote profusion of active tissues. Catecholamines also stimulate the renin-angiotensin-aldosterone system to secrete aldosterone and thus to protect against hypohydration; they work in the pancreas to decrease insulin and increase glucagon to increase lipolysis and maintain blood glucose levels for working muscles. As in the case of the other adrenal hormones, increased secretion is directly related to exercise intensity. However, unlike many other hormones discussed thus far, basal levels of catecholamines do change with prolonged training.[134] Findings have reported a higher adrenaline response to exercise in endurance-trained compared with untrained subjects in response to intense exercise at the same relative intensity as all-out exercise. This phenomenon is referred to as the "sports adrenal medulla." This higher capacity to secrete adrenaline was observed both in response to physical exercise and to other stimuli such as hypoglycemia and hypoxia.[135] Long-term endurance training can enhance plasma catecholamine concentrations in response to supramaximal exercise.[134]

Gonadal Hormones

Testosterone is a male sex hormone produced in the testes; it has both androgenic and anabolic effects. Testosterone has an important role as a regulator of muscle protein synthesis and hypertrophy, so its importance for exercise is clear. Regulation of testicular production occurs via a negative feedback loop involving the anterior pituitary, hypothalamus, and testicles, referred to as the *hypothalamic-pituitary-testicular* (HPT) *axis*. High-intensity, near-maximal resistance training will increase systemic testosterone levels.[136] This increased testosterone level as a result of resistance training appears to become blunted in persons who are long-term weight lifters.[137] Short-term, moderate intensity and low-volume endurance training can also significantly increase testosterone concentration in previously untrained men.[138] Interestingly, prolonged endurance training may lead to decreased testosterone levels. The mechanism of this lowering is currently unclear but may be related to dysfunction within the HPT axis brought about by months or years of endurance training.

Estrogen and progesterone are the female sex steroid analogs produced in the ovary; they are also regulated via a negative feedback loop involving the pituitary, hypothalamus, and ovaries. Resistance exercise increases levels of both estrogen and progesterone,[138] which have been shown to increase muscular strength in women. Evidence indicates that estradiol working through estrogen receptors does not accomplish this increase in muscular strength by affecting muscle size but rather by improving the intrinsic quality of skeletal muscle, whereby fibers are enabled to generate more force.[139] Estrogen is also implicated in bone health, and estrogen's effects on bone would be enhanced if, in addition to the direct effects of the hormone on the skeleton, muscle contractions were more forceful and/or numerous, creating additional osteogenic stimuli. Although estrogen may have a beneficial effect with resistance training in women, prolonged endurance training in the face of calorie restriction may create an energy deficit, which then leads to a lowering of the estrogen level and a decreased protective effect on bone. Women with menstrual disturbances who have low estradiol and progesterone levels have an attenuated anabolic hormone response to exercise, suggesting that menstrual disorders may affect exercise-induced changes to strength and bone health in women.[140]

Pancreatic Hormones

Insulin and glucagon are secreted by the pancreas and interact with muscle, liver, and fat cells. Insulin causes muscles and the liver to take up glucose from the blood stream and store it as glycogen while inhibiting the use of fat as a fuel source by inhibiting the secretion of glucagon. Glucagon, meanwhile, acts to raise blood sugar levels by signaling the liver to break stored glycogen down into available glucose. Thus insulin and glucagon are part of a feedback system that regulates blood glucose levels and as such are important hormones for exercise.

During a bout of exercise, the working muscle rapidly increases its need for glucose. Therefore it does not result in hypoglycemia. This augmentation of glucose utilization must be accompanied by an almost synchronous and equivalent increase in glucose production.[141] Glucagon responds rapidly to meet this need; simultaneously, a decrease in plasma insulin levels during exercise occurs to maintain higher levels of blood glucose. Glucagon also plays a critical role in coupling the glycogenolysis response of the liver to the glucose demands of the working muscle.

A single bout of exercise increases skeletal muscle glucose uptake via an insulin-independent mechanism that bypasses the typical insulin signaling defects associated with these conditions. However, this “insulin sensitizing” effect is short-lived and disappears after approximately 48 hours. In contrast, some studies demonstrate that prolonged exercise training results in a persistent increase in insulin action in skeletal muscle.[142] Other studies, however, suggest that there is no attenuation period of insulin sensitivity in obese persons, implicating exercise as much more effective than diet in changing body composition among those who are obese.[143,144] Finally, it has been shown that insulin plays only a small role in actual muscle protein synthesis but may inhibit muscle protein breakdown, which, combined with the ingestion of small amounts of protein and carbohydrate, can transiently increase muscle protein buildup.[145] Because of insulin’s ability to direct glucose into muscle tissue, it could be useful as an anabolic agent and thus may have the potential for abuse in an effort to maximize muscle gains from exercise or increase glycogen storage for prolonged endurance events.[146]

EXERCISE CAPACITY

Peak exercise capacity is the maximum ability of the cardiovascular system to deliver oxygen to exercising skeletal muscle and the exercising muscle’s ability to extract oxygen from blood. Maximum oxygen uptake, or Vo_2 max, is generally used to provide an overall assessment of exercise capacity (Fig. 6.5). The Fick equation states that oxygen uptake (Vo_2) equals cardiac output (CO) times the arterial oxygen content minus the mixed venous oxygen content: $Vo_2 = CO \times (CaO_2 - CvO_2)$, where CO equals stroke volume (SV) times HR; therefore $Vo_2 = (SV \times HR) \times (CaO_2 - CvO_2)$. At maximal exercise, Vo_2 max is calculated as follows: $(SVmax \times HRmax) \times (CaO_2max - CvO_2\ max)$, which reflects a person’s ability to take in, transport, and utilize oxygen.

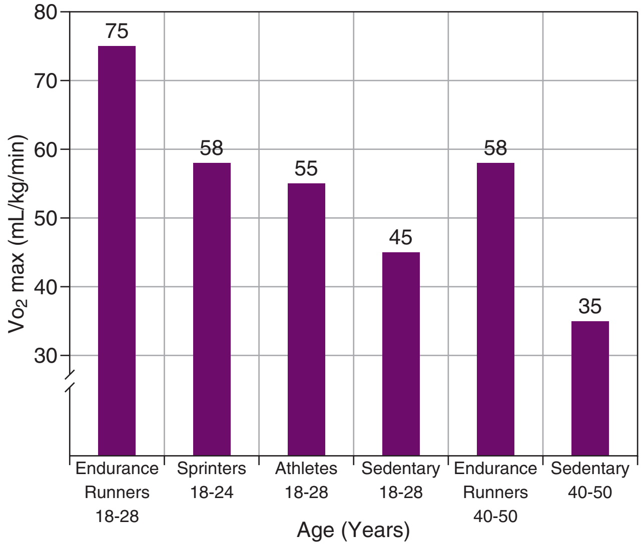

Fig. 6.5 Contrast of maximum oxygen uptake *(Vo_2 max)* among men of various ages and aerobic fitness levels. (Data from Pollock ML, Wilmore JH, Fox SM. *Health and Fitness Through Physical Activity*. New York: John Wiley and Sons; 1978.)

Vo_2 max has become the preferred laboratory measure of cardiorespiratory fitness and is the most important measurement during functional exercise testing. In healthy people, a Vo_2 plateau occurs at near-maximal exercise. This plateau in Vo_2 has traditionally been used as the best evidence of Vo_2 max.[146] It represents the maximal achievable level of oxidative metabolism involving large muscle groups. However, in clinical testing, a clear plateau may not be achieved before symptoms limit exercise. Consequently, peak Vo_2 is often used as an estimate of Vo_2 max.[96]

In healthy people, the Vo_2 max response to exercise is linear until Vo_2 max is achieved. Exercise training, however, will enable the person to increase his or her Vo_2 max. Training leads to a lower resting HR but does not significantly affect maximal HR. Initially the SV increases with exercise, but then the response is attenuated. Training increases resting SV and therefore SV at each level of work. Finally, the difference between the arterial and venous oxygen content widens with training. Thus although Vo_2 max increases with training, an age-related decline occurs. Aging is associated with a progressive decline in the capacity for physical activity, mainly as a result of this reduction in Vo_2 max. A reduction in muscle oxygen delivery, principally due to reduced CO, appears to play the dominant role up until late middle age. Mitochondrial dysfunction is the major factor in older age because skeletal muscle Vo_2 max declines by approximately 50% compared with younger muscle.[147]

The ventilator anaerobic threshold, formerly referred to as the anaerobic threshold, is another index used to estimate exercise capacity. During the aerobic phase of exercise (up to 60% of max Vo_2), ventilation increases linearly with Vo_2 and mirrors carbon dioxide produced aerobically in the muscles. No significant change in blood lactate levels occurs during this phase because muscle lactic acid production is minimal. As the intensity and duration of exercise continue, the oxygen supply can no longer keep up with the increasing metabolic requirements of the exercising muscles.

At this time, lactic acid and therefore blood lactate concentration increases. The Vo_2 at the onset of this blood lactate spike is referred to as the *lactate threshold* or the *ventilator anaerobic threshold*. This threshold also denotes the point at which minute ventilation increases disproportionately relative to Vo_2, which generally occurs at 60% to 70% of Vo_2 max.

Cardiorespiratory Response to Exercise

The cardiovascular system serves several important functions during exercise. It delivers oxygen to working muscles, oxygenates blood by returning it to the lungs, delivers nutrients and fuel to active muscles, transports hormones, and transports heat from the core to the skin. Exercise places an increased demand on the cardiovascular system. Oxygen demand by the muscles increases sharply, metabolic processes speed up, and more waste is created, more nutrients are utilized, and body temperature rises.[148] To perform efficiently and effectively, the cardiovascular system must regulate these changes to meet the body's increased demands.

Exercise CO is defined as HR multiplied by SV, with SV defined as the volume of blood ejected per ventricular contraction. Aerobic capacity is determined by the ability of the body to modify the CO in response to exercise, with approximately a fivefold increase in output during vigorous exercise. This increase in CO is primarily due to the increase in HR with exercise; to a lesser extent, it is due to an increase in SV. The resting HR is typically between 60 and 100 beats per minute and increases linearly with exercise intensity to a maximal HR of 200 beats per minute in the young adult (Fig. 6.6).[149] Individual maximal HR decreases with age and may be loosely approximated by 220 minus age (in years).

Endurance training is associated with an increased CO and volume load on the left and right ventricles, causing the endurance-trained heart to generate dilation of the left ventricle combined with a mild to moderate increase in left ventricular wall thickness. This training-induced increase in CO allows trained athletes to have a lower resting HR compared with persons who are not trained. Also, during exercise, the active muscles of the lower extremities require increased blood flow; therefore peripheral vascular resistance decreases to accommodate this need. To generate this large CO, athletes must increase their SV to counteract the decrease in HR and vascular resistance. Endurance athletes have larger increases in left ventricular end-diastolic volume compared with nonathletes, which allows them to generate the necessary larger SV. SV increases early in exercise, but unlike HR, it plateaus before maximal intensity at about 40% to 60% of Vo_2 max.[150] Trained athletes have also demonstrated the ability of the heart to adapt to maintain CO. Training-related expansion of vascular volume is associated with decreased HR response to baroreceptor stimulation.[151] Levels of skeletal muscle blood flow increase up to 20 times resting value during exercise to maximize oxygen delivery.[152] Visceral vasoconstriction likewise aids in redistributing blood flow to the skeletal muscle (Table 6.3).

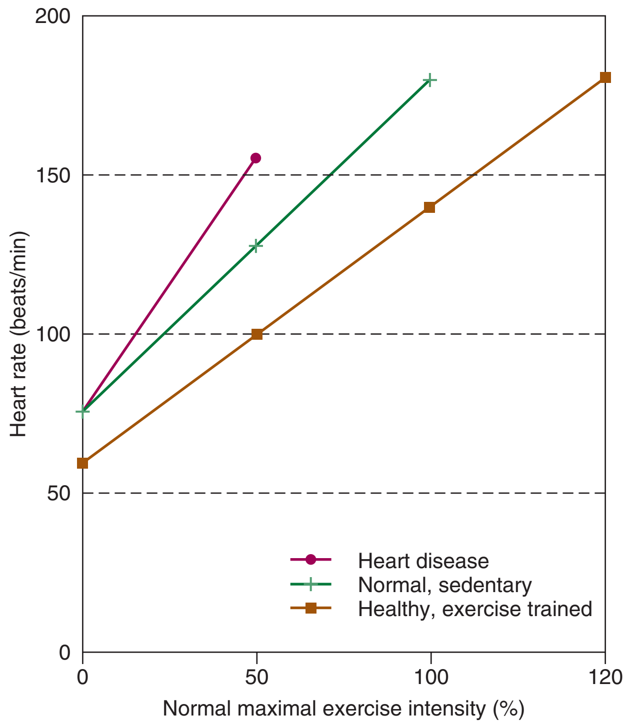

Fig. 6.6 Heart rate increase in proportion to exercise intensity among groups in various states of health. (Data from Hasson S. *Clinical Exercise Physiology*. St. Louis: Mosby; 1994.)

TABLE 6.3 Cardiovascular Response to Exercise

Function	Response
Cardiac output	Large increase
Heart rate	Large increase (primarily responsible for cardiac output increase)
Stroke volume	Modest increase
Systolic blood pressure	Modest increase
Pulse pressure	Modest increase
Mean blood pressure	Modest increase
Total peripheral resistance	Large decrease (massive skeletal muscle vasodilation dominates over visceral vasoconstriction)
Arteriovenous oxygen difference	Increase due to oxygen consumption

Respiratory Response

The lungs are critical in oxygenating blood and in their ability to maintain consistent arterial oxygen concentrations both at rest and with the increased demands of exercise. The ventilatory demands of vigorous exercise require airway flow rates that often exceed 10 times resting levels and tidal volumes that approach five times resting levels.

Basic to understanding this respiratory response is the importance of diffusion in the process of gas exchange between alveolar air and pulmonary capillary blood. For gas to be exchanged, concentration differences must be maintained by the ventilation of lung airway and the perfusion of pulmonary capillaries. This diffusion rate is directly rated to the concentration gradient of oxygen and the surface area of lung available for diffusion and is the most important factor for lung diffusion; therefore it is necessary to ensure that oxygen and carbon dioxide gradients are sufficiently steep to maximize diffusion.

It was long thought that the capacity of the healthy respiratory system is overbuilt for the demands placed on ventilation and gas exchange by exercise.[153] However, for endurance athletes, the pulmonary system may lag behind the exceptional cardiovascular and aerobic muscular adaptations.[154] This potential for ventilation-perfusion inequality during high-intensity exercise may compromise arterial saturation and the capacity for oxygen transport; it is termed "exercise-induced arterial hypoxemia." The exact etiology of these respiratory system limitations to exercise is unknown but may be due to the specific ability of training to increase the capacity of the muscles and heart for oxygen transport and utilization without significantly changing the structural or functional capacities of the lungs and airways.[155]

CONCLUSION

Exercise physiology—the study of the identification and study of physiologic mechanisms underlying physical activity and the body's response to exercise—provides the scientific basis for training that is used to maximize performance. A better understanding of the underlying mechanisms of muscle structure, function, and adaptation will aid providers in the evaluation and treatment of exercising athletes.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:
Brooks GA. Cell-cell and intracellular lactate shuttles. *J Physiol*. 2009;587(23):5591–5600.

Level of Evidence:
II

Summary:
This article contradicts previously held theory that lactate is a waste product in muscle metabolism, indicating that it is a viable substrate for muscle energy and that glycolytic and oxidative pathways work in concert rather than in competition.

Citation:
American College of Sports Medicine. *ACSM's Guidelines for Exercise Testing and Prescription*. 7th ed. Philadelphia: Lippincott, Williams & Wilkins; 2006:237–251.

Level of Evidence:
III

Summary:
A comprehensive guide to exercise testing and prescriptions for athletes of all ages and stages of their careers.

Citation:
Herzog W, Duvall M, Leonard TR. Molecular mechanisms of muscle force regulation: a role for titin? *Exerc Sport Sci Rev*. 2012;40(1):50–57.

Level of Evidence:
II

Summary:
The authors support their theory that in addition to the proteins actin and myosin discussed in the cross-bridge theory, a new protein may have a role in muscle force. They demonstrate that titin plays a major role in muscle force regulation, particularly for eccentric concentrations and at long muscle and sarcomere lengths.

Citation:
Reid KF, Callahan DM, Carabello RJ, et al. Muscle power failure in mobility limited older adults: preserved single fiber function despite lower whole muscle size, quality and neuromuscular activation. *Eur J Appl Physiol*. 2012;112(6):2289–2301.

Level of Evidence:
II

Summary:
This study contradicts the previously held theory that muscle fibers degenerate as we age. The authors find that in older adults, contractile properties of muscle fibers are preserved in an attempt to maintain overall muscle function.

Citation:
Yan Z, Okutsu M, Akhtar YN, et al. Regulation of exercise-induced fiber type transformation, mitochondrial biogenesis, and angiogenesis in skeletal muscle. *J Appl Physiol*. 2011;110:264–274.

Level of Evidence:
III

Summary:
This article summarizes the variety of physiologic and biochemical adaptions in skeletal muscle, which forms the basis for the improvement of physical performance and other health benefits.

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